

Mathematical Models at the Population Level to Support Immunotherapy Design

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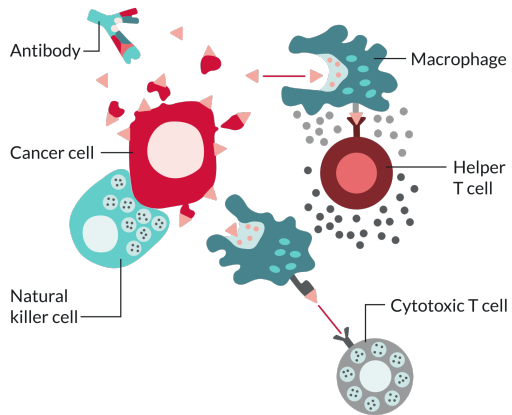
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1. Immune System Functioning

Immune System Functioning



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2. Formulation of the Model

Foundational ODE Model

The model developed by Simmons and Levy [?] describes the development of cellular exhaustion. It uses five state variables:

- N : Tumor volume.
- K : Residual healthy CTL (CD8+) population.
- B : Progenitor Exhausted T cells.
- E : Terminally Exhausted T cells.
- P : Interleukin-2 (IL-2) cytokine concentration.

ODE Model (Equations)

The dynamics of the system are described by the following set of ODEs:

$$\begin{cases} \dot{N}(t) &= \alpha N \left(1 - \frac{N}{T_c}\right) - h \left(\frac{\lambda_K N K + \lambda_E N E}{N + K + E + 1}\right) \\ \dot{K}(t) &= h P K - \delta_K K \\ \dot{B}(t) &= 2^m s_B N + h P B - r_E B - \delta_B B \\ \dot{E}(t) &= r_E B - \delta_E E \\ \dot{P}(t) &= \rho_K K + \rho_B B - h P (K + B) - \delta_P P \end{cases}$$

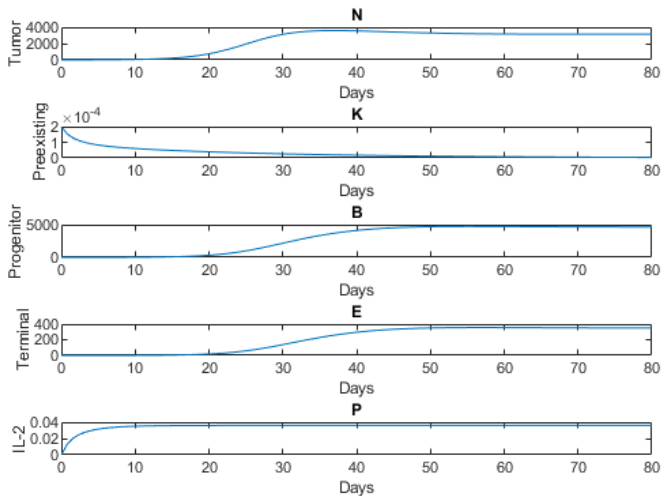


Figure: Reproduction of the model dynamics showing the evolution of N , K, B, E, and P over time.

3. Extending the Model

Extending the model: multiple exhaustion states

To capture the heterogeneity of T cell exhaustion, the E group of cells has been divided following different levels of exhaustion:

- E_0 : Initially Exhausted T cells (functional, capable of tumor killing).
- $E_{1/2}$: Middle-exhausted T cells (partially dysfunctional).
- E_1 : Deeply-exhausted T cells (fully exhausted, negligible killing capabilities).

ODE Model with Multiple Exhaustion States

The extended model is described by the following set of ODEs:

$$\left\{ \begin{array}{l} \dot{N}(t) = \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k N K + \lambda_0 N E_0 + \lambda_{1/2} N E_{1/2} + \lambda_1 N E_1}{N + K + E_0 + E_{1/2} + E_1} \right), \\ \dot{K}(t) = h P K - \delta_k K, \\ \dot{B}(t) = 2^m S_B N + h P B - r_E B - \delta_B B, \\ \dot{E}_0(t) = r_E B - r_t E_0 - \delta_0 E_0, \\ \dot{E}_{1/2}(t) = r_t E_0 - r_t E_{1/2} - \delta_{1/2} E_{1/2}, \\ \dot{E}_1(t) = r_t E_{1/2} - \delta_1 E_1 \\ \dot{P}(t) = \rho_K K + \rho_B B - h P (K + B) - \delta_P P, \end{array} \right. \quad (1)$$

Homogeneity in killing and death rates

With the assumption of homogeneity in the death rates ($\delta_0 = \delta_{1/2} = \delta_1 = \delta_E$) and in the killing rates ($\lambda_0 = \lambda_{1/2} = \lambda_1 = \lambda_E$), we can recover the original equation for E :

$$\dot{E} = \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1$$

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$$\begin{aligned}\dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ &= (r_E B - r_t E_0 - \delta_E E_0) + (r_t E_0 - r_t E_{1/2} - \delta_E E_{1/2}) + (r_t E_{1/2} - \delta_E E_1)\end{aligned}$$

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$$\begin{aligned}\dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ &= (r_E B - \cancel{r_t E_0} - \delta_E E_0) + (\cancel{r_t E_0} - \cancel{r_t E_{1/2}} - \delta_E E_{1/2}) + (\cancel{r_t E_{1/2}} - \delta_E E_1)\end{aligned}$$

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$$\begin{aligned}\dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ &= (r_E B - \cancel{r_t E_0} - \delta_E E_0) + (\cancel{r_t E_0} - \cancel{r_t E_{1/2}} - \delta_E E_{1/2}) + (\cancel{r_t E_{1/2}} - \delta_E E_1) \\ &= r_E B - \delta_E (E_0 + E_{1/2} + E_1)\end{aligned}$$

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$$\begin{aligned}\dot{E} &= \dot{E}_0 + \dot{E}_{1/2} + \dot{E}_1 \\ &= (r_E B - \cancel{r_t E_0} - \delta_E E_0) + (\cancel{r_t E_0} - \cancel{r_t E_{1/2}} - \delta_E E_{1/2}) + (\cancel{r_t E_{1/2}} - \delta_E E_1) \\ &= r_E B - \delta_E (E_0 + E_{1/2} + E_1) \\ &= r_E B - \delta_E E.\end{aligned}$$

Homogeneity in killing and death rates

In the same way, tumor equation is the same as the original one:

$$\begin{aligned}\dot{N} &= \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k N K + \lambda_0 N E_0 + \lambda_{1/2} N E_{1/2} + \lambda_1 N E_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\ &= \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k N K + \lambda_E N E_0 + \lambda_E N E_{1/2} + \lambda_E N E_1}{N + K + E_0 + E_{1/2} + E_1 + 1} \right) \\ &= \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k N K + \lambda_E N (E_0 + E_{1/2} + E_1)}{N + K + (E_0 + E_{1/2} + E_1) + 1} \right) \\ &= \alpha N \left(1 - \frac{N}{T_C}\right) - h \left(\frac{\lambda_k N K + \lambda_E N E}{N + K + E + 1} \right)\end{aligned}\tag{2}$$

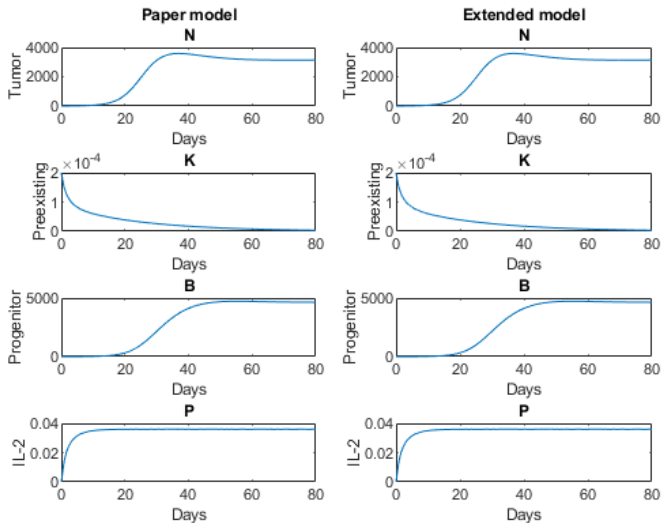


Figure: Comparison between the original system and the new one.

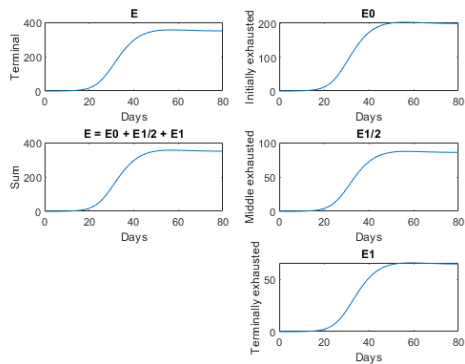
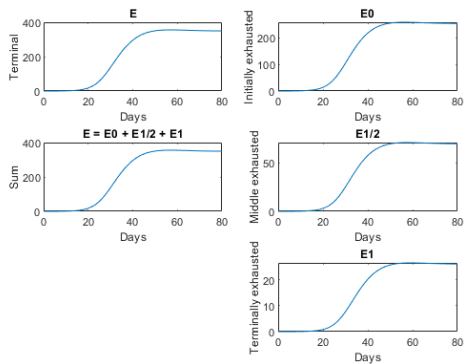


Figure: Comparison between the original system and the new one with $r_t = 5r_E$ on the left and $r_t = 10r_E$ to the right.

Heterogeneity in killing and death rates

- Deeper exhaustion is linked with a shorter lifespan:

$$\delta_t(y) = \delta_{E_0} + (\delta_{E_1} - \delta_{E_0})y$$

- Progressive exhaustion severely affects effector function:

$$\delta_t(y) = \delta_{E_0} + (\delta_{E_1} - \delta_{E_0})y$$

- T cells progressively pass from one state to the other in the following order:

$$E_0 \rightarrow E_{1/2} \rightarrow E_1$$

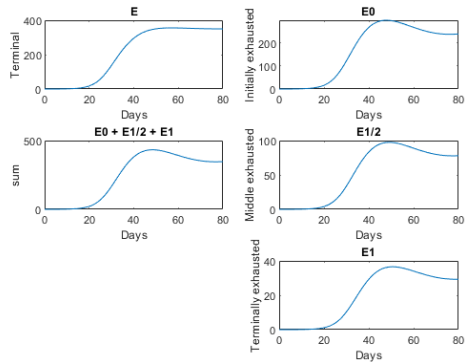
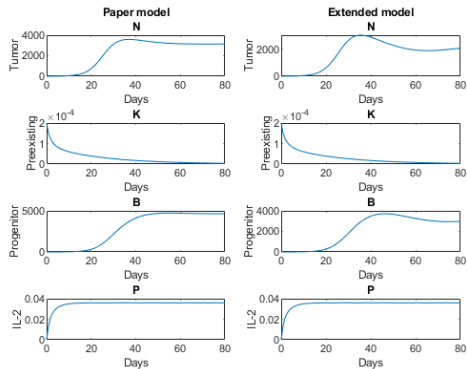


Figure: Comparison between the original system and the new one with $r_t = 5r_E$ and heterogeneous killing and death rates.

4. Continuous Population Model
